

FORT-DE-FRANCE, Martinique

WMO: 789250

Lat: 14.595N

Lon: 60.996W

Elev: 8

StdP: 101.23

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00404

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.9	21.0	17.1	12.3	24.4	18.0	12.9	24.8	10.4	27.7	9.9	27.8	1.3	100	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	32.2	25.8	32.0	25.7	31.3	25.6	26.9	30.6	26.6	30.3	26.4	30.0	5.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	29.1	25.6	20.8	28.7	25.2	20.3	28.4	84.7	30.7	83.4	30.6	82.2	30.1	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.7	8.1	DB	17.9	33.6	1.0	0.6	17.1	34.1	16.5	34.5	16.0	34.8	15.2	35.3	(4)
(5)			WB	17.1	27.8	0.9	0.6	16.5	28.2	15.9	28.6	15.4	29.0	14.8	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	25.6	25.5	25.9	26.6	27.5	27.9	27.9	28.0	27.9	27.5	27.0	26.2	(6)
(7)		DBStd	1.25	0.85	0.90	0.98	0.96	0.87	0.75	0.76	0.82	0.83	0.86	0.88	0.88	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6190	484	435	493	498	542	536	555	557	536	544	509	502	(10)
(11)		CDD18.3	3148	226	202	234	248	283	286	297	298	286	286	259	243	(11)
(12)		CDH23.3	30037	1654	1467	1857	2239	2911	3113	3246	3244	3026	2852	2393	2036	(12)
(13)	CDH26.7	9006	304	287	448	632	908	961	1034	1137	1136	995	707	458	(13)	
(14)	Wind	WSAvg	4.0	4.3	4.6	4.3	4.2	4.3	5.0	4.7	3.7	3.0	3.2	4.0	(14)	
(15)	Precipitation	PrecAvg	2225	127	85	95	116	154	198	247	299	238	239	247	173	(15)
(16)		PrecMax	3055	198	198	186	419	368	342	379	783	413	474	554	375	(16)
(17)		PrecMin	1759	52	10	36	32	51	95	103	157	101	94	57	76	(17)
(18)		PrecStd	372	39	35	42	92	90	67	74	124	82	104	118	67	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	30.7	31.2	32.0	32.2	32.1	32.0	32.6	33.2	33.0	32.0	30.8	(19)
(20)			MCWB	23.7	23.9	24.0	24.9	25.5	25.9	26.0	26.3	26.0	26.0	25.8	24.8	(20)
(21)		2%	DB	29.2	29.5	30.2	31.0	31.2	31.2	31.2	32.0	32.2	32.0	31.1	30.0	(21)
(22)			MCWB	23.5	23.2	23.6	24.6	25.1	25.4	25.6	26.0	25.8	25.8	25.5	24.5	(22)
(23)		5%	DB	28.9	29.0	29.6	30.2	31.0	30.9	31.0	31.2	31.9	31.2	30.6	29.3	(23)
(24)			MCWB	23.4	23.0	23.3	24.0	24.9	25.3	25.6	25.8	25.8	25.7	25.4	24.1	(24)
(25)		10%	DB	28.2	28.2	29.0	29.8	30.2	30.2	30.4	31.0	31.2	30.9	30.0	29.0	(25)
(26)			MCWB	23.1	22.7	23.1	23.9	24.7	25.1	25.4	25.8	25.8	25.7	25.2	24.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.0	24.7	25.1	25.8	26.7	26.9	27.0	27.2	27.2	27.2	26.7	26.0	(27)
(28)			MCD B	28.2	28.8	29.2	29.9	30.4	30.3	30.4	30.7	31.1	30.6	30.3	29.1	(28)
(29)		2%	WB	24.5	24.1	24.4	25.2	26.1	26.4	26.6	26.7	26.7	26.7	26.3	25.5	(29)
(30)			MCD B	27.7	28.0	28.6	29.2	29.8	29.8	29.9	30.4	30.6	30.3	29.8	28.7	(30)
(31)		5%	WB	24.0	23.7	24.0	24.8	25.6	26.1	26.2	26.5	26.5	26.4	25.9	25.0	(31)
(32)			MCD B	27.4	27.4	28.1	28.7	29.3	29.3	29.5	30.0	30.4	30.0	29.4	28.2	(32)
(33)		10%	WB	23.7	23.3	23.6	24.4	25.2	25.7	25.9	26.2	26.2	26.0	25.6	24.6	(33)
(34)			MCD B	27.0	27.0	27.6	28.2	28.9	28.9	29.2	29.6	30.0	29.7	28.8	27.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	6.4	6.7	6.7	6.1	5.3	5.5	6.3	7.3	7.1	6.6	6.2	(35)
(36)			MCDBR	6.5	7.0	7.4	7.3	6.7	5.7	5.8	6.8	8.1	7.7	7.3	6.6	(36)
(37)			MCWBR	2.9	2.9	2.9	2.7	2.5	2.3	2.4	2.7	3.1	3.2	3.1	3.0	(37)
(38)		5% WB	MCDBR	5.9	6.3	6.6	6.8	6.2	5.4	5.5	6.5	7.3	7.2	6.7	6.2	(38)
(39)			MCWBR	2.9	2.9	2.9	2.8	2.6	2.5	2.6	2.9	3.2	3.3	3.2	3.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.366	0.381	0.380	0.416	0.464	0.550	0.517	0.462	0.424	0.385	0.377	0.376	(40)	
(41)		taud	2.607	2.539	2.549	2.469	2.323	2.038	2.144	2.358	2.490	2.625	2.630	2.601	(41)	
(42)		Ebn at Noon	932	929	934	893	838	760	788	841	881	915	915	912	(42)	
(43)		Edh at Noon	94	105	106	115	130	171	155	126	111	95	91	93	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.13	5.82	6.29	6.44	6.26	6.07	6.21	6.28	5.96	5.39	4.95	4.85	(44)	
(45)		RadStd	0.15	0.13	0.27	0.28	0.40	0.30	0.27	0.30	0.27	0.27	0.36	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.30	+0.68	N/A	+0.35	N/A	N/A	N/A	N/A	+113	+112	(46)	
(47)	Regional (2 neighbors)	N/A	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	+113	+112	(47)	

Nomenclature: See separate page